Quick selection of commands for GTEK programmers
L - formatted list command, use start/end addresses for other than full eprom.
M - gives menu with a cr, or selection with proper letter (me = 2764)
OI- outputs intel hex file, use start/end addresses for other than full eprom.
OM- outputs motorola hex file. "   "
OT- outputs Tektronic hex file. "   "
P - program ascii hex characters, use start address if not at beginning.
R - read ascii hex characters, use start/end address if not full eprom.
T - toggle command:
    TB- toggle byte... toggle high/low byte when used with split eproms.
    TC- toggle compare mode.
    TI- toggle intelligent programming mode.
    TN- toggle checksum. may use start/end addresses.
    TR- toggle reset. resets all toggles.
    TS- toggle split mode. used to program 16 bit split eproms.
    TZ- toggle zap mode. used to erase some EE proms.
V - verify blank. use start/end addresses if necessary.
X - version command. gives version of programmer operating system.
$ - terminate any programmer operation.

examples of commands:
L ff   lists contents of eprom in a formatted ascii hex dump from 0-ff hex.
l0-ff   same results.
L0000-00ff   same results.
OI3000-3FFF   output an Intel Hex file from locations 3000 to 3FFF hex in an eprom. You must realize that if you have a 2732 selected for instance, you will actually get the contents of locations 0000 through 0fff hex. If you had a 2764 selected, you would actually get the contents of 1000-1fff hex. If you had a 27128 selected you would actually get the contents of 3000-3fff hex. The addresses are only significant to the programmer in relation to the eprom size.
TN(cr)   This causes the programmer to give you a 16 bit checksum (addition of all 8 bit bytes without a 16 bit carry) of the entire eprom selected. An empty 2764 gives a checksum of E000h.
TN ff   gives the checksum of 0000 through 00ff hex in an eprom.
P1000 ffff$ programs the eprom selected at locations 1000h and 1001h with the data ff and ff. Programmer returns to the command mode.

If you are not using our interface program, PGX, then the best way to use a modem program to capture data is to give the programmer a pending command that the next character would be a (cr) for execution. Then give the modem

program the "capture incoming data" command to open your buffer. When you begin communication with the programmer again, send a (cr) to the programmer
to begin the execution of the pending command. The (cr) will get put into the buffer that you opened, but if you are going to send it back to the prog-
rammer, it doesn't cause any problems. A (cr) causes the prompter to be
reissued. You may also edit the file with a word processor like Wordstar in
the Non-Document mode to remove the extraneous characters from the file. Make
sure that you don't use any word processor that will change the file to any
other kind of format. The programmer will promptly start telling you *SN err
@ 0000 three times then appear to hang up. Any characters from the command
mode that cause three syntax errors in a row will cause the programmer to go
looking for a new baud rate. You must abort sending the file to the programmer
and then send the programmer about 4 to 6 spaces to cause it to lock on to the
baud rate again.

Wires necessary in RS-232 cable to communicate with GTEK programmers.

```
GTEK side (female)                          to      DTE (IBM PC)

EG--1--------(optional but hooked in programmer)----1-EG (earth ground)
TXD-2-----------------------------------------------3-RXD
RXD-3-----------------------------------------------2-TXD
RTS-4---------(possibly optional)-------------------6-DSR (Data Set Ready)
CTS-5----------------------------------------------20-DTR (Data Terminal Ready)
DSR-6---------(possibly optional)-------------------4-RTS (Ready To Send)
SG--7-----------------------------------------------7-SG (signal ground)
DTR-20----------------------------------------------5-CTS (Clear To Send)
```

you may also jumper pin 6 DSR to pin 8 CD at both ends. This may
be necessary if you use an IBM PC and other types of software.